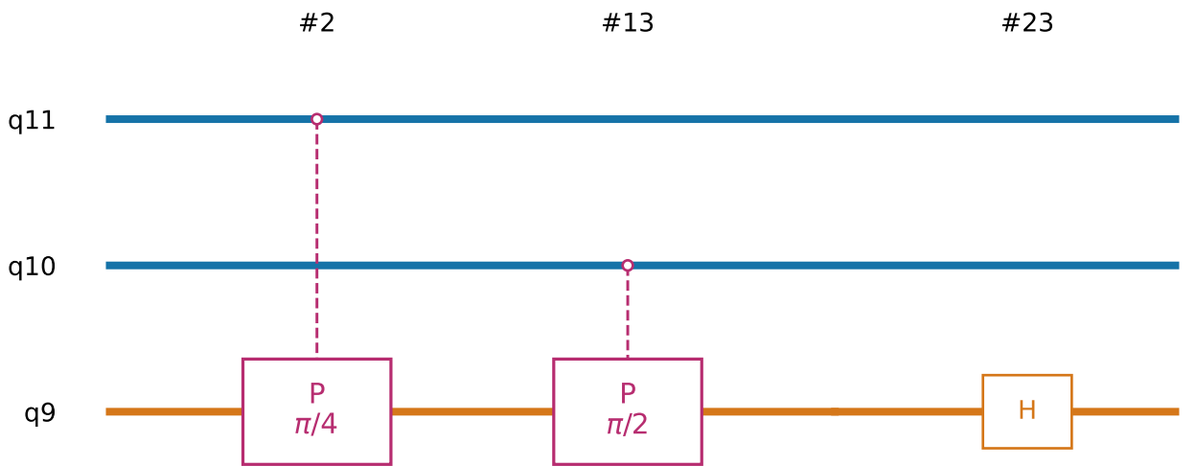
QFT12: what the first SA version improves

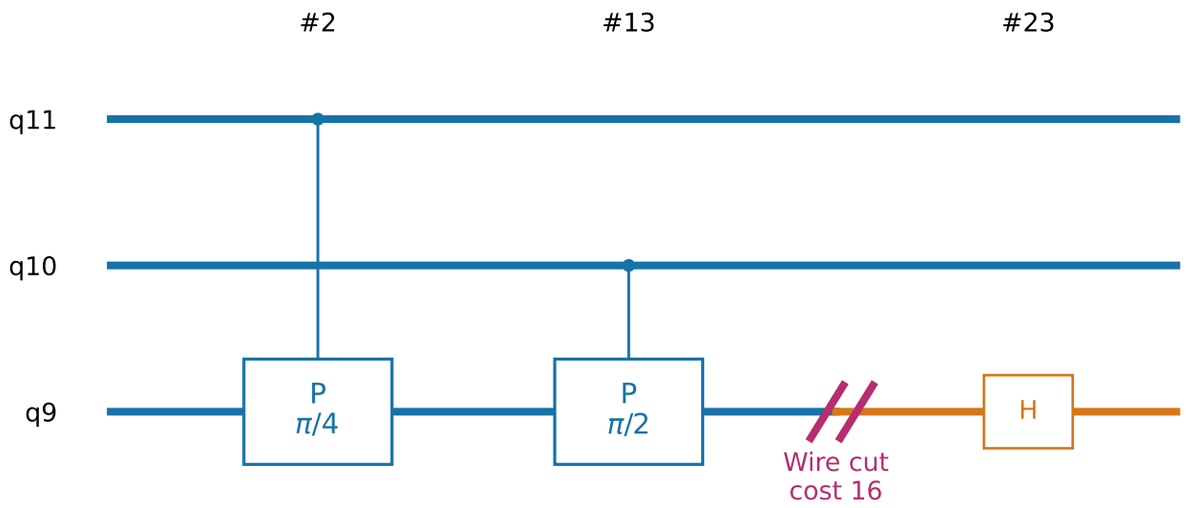
Spatial seed: 796,655.92 → Hybrid: 701,729.05 | 11.92% lower overhead

Spatial: cut both early gates



Inset omits other gates; original operation indices shown.

Hybrid: keep both early gates local



Inset omits other gates; original operation indices shown.

Replace two gate-cut factors 3.1165 × 5.8284 = 18.1644 with one wire-cut factor 16.
The other 30 gate cuts are unchanged; no SWAP cuts or excisions.

A = q0, q1, q10, q11, early q9 (5 qubits)
B = q2–q8, late q9 (8 qubits); both within limit 9

Quality limit: best found in 1/50 final starts; median overhead 34.6 million.
Reconstructed benchmark; no optimality certificate or hardware execution.